

Week 5: Daily Journals

Monday:

I started the week by working out a GitHub issue with Jonah. The log files created every 30 seconds were causing issues because some files were over 100 MB and not uploading making the repository and my directory did not line up. We resolved the issue after lunch, then I spent some time troubleshooting the 630 nm camera issue as well as the issues I had with the X-Ray Sensor.

Tuesday:

I started the day by running the X-Ray sensor with the X-Ray sensor script, and I was able to get understandable data from the sensor. After I showed that the X-Ray sensor worked reliably, my team gave a mini presentation to one of the summer camp groups visiting SFU. Then, we were able to start testing with the switchboard, however on the board there is an error where the power and GPIO connector is expecting a male connector instead of a female connector. So, we desoldered and resoldered the connectors, then we wrapped up for the day.

Wednesday:

For today, our main priority is making sure the power cable and the GPIO cable were verified. Before that, we twisted the GPIO cable wires together into bundles to eliminate any interference. Then we moved onto testing the power connector which had no issues after testing, but when we tested the GPIO cable, there was a problem with the LED lights on the switchboard, so we decided to fix the resistors so the LED lights would shine brighter.

Thursday:

Today, we started by still trying to fix the switchboard. We were having trouble desoldering one of the resistors, but eventually we thought we got it. However, whenever we tried to test the GPIO lines, we ran into some issues. We thought the issue was with the switchboard and that it was unfixable, so we restarted the switchboard.

Friday:

Today, we finished remaking the switchboard, but whenever we tested it with the X-Ray sensor, we did not get any results. We came to the conclusion that the switchboard was not the issue, but there was an issue with the X-Ray sensor. We also determined that there is a crosstalk happening in the GPIO cable. We decided to remove any unnecessary connections from the GPIO cable in an attempt to remove crosstalk. Next, we made a new 9-pin connector so that we could work around the switchboard. We got the X-Ray sensor to work without the switchboard, so our next steps are to test the X-Ray sensor while using the switchboard.

Pictures



Figure 2: X-Ray Cable

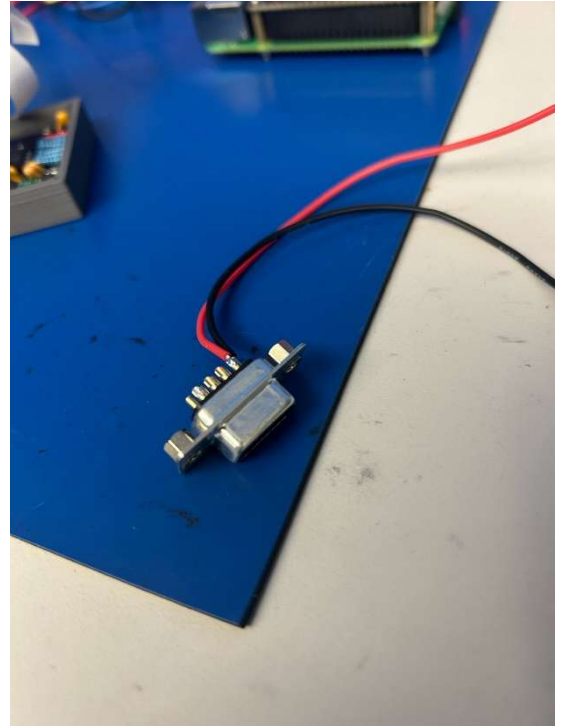


Figure 1: SDR Cable